

SHOULD-COST ANALYSIS REPORT

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Bottom-Up Manufacturing Cost Model · Fully Traceable Rate Data

Bosch IPB2.0 ECU PCBA

Commodity: PCBA · Currency: GBP · FX Rate: 1.0000 to GBP · Operations: 4 · Region: CN

Total Should-Cost	Material	Process	Labour	Margin
£185.32	72.1%	11.0%	1.1%	7.4%

Model Confidence: **Low** · 4 traced operations · 9 data points auditable

Key Assumptions

Annual volume: 150,000 · Region: CN · Commodity: pcba
Alloy / material: Virtual / Pass-through · Net weight: 0.000 kg
Material utilisation: 100% · Overhead: 9% of factory base · Margin: 8% of subtotal · Operations: 4

§1 — 8-Bucket Cost Breakdown

Cost Bucket	Amount (GBP)	% of Total	Cost Mix Bar
1. Raw Material	£133.56	72.1%	
2. Process (Machine)	£20.38	11.0%	
3. Direct Labour	£1.99	1.1%	
4. Tooling (amortised)	£0.31	0.2%	
5. Packaging	£0.73	0.4%	
6. Logistics	£0.57	0.3%	
Factory Cost	£157.53	85.0%	
7. Overhead (SG&A)	£14.06	7.6%	
Subtotal	£171.59	92.6%	
8. Supplier Margin	£13.73	7.4%	

Cost Bucket	Amount (GBP)	% of Total	Cost Mix Bar
TOTAL SHOULD-COST	£185.32	100.0%	

§2 — Commercial Parameters

Overhead Rate	9.0%	Supplier Margin	8.0%
Packaging / part	£0.73	Logistics / part	£0.57
Total Tooling Cost	£46000.00	Amortisation Volume	150,000 parts

S3 — Material Detail

Parameter	Value	Unit	Notes
Material ID	mat-virtual	ID	
Grade / Specification	Virtual / Pass-through		Placeholder for directCost modules (painting, BIW, PCB, PCBA). Price is irrelevant — directCost overrides. Regional adj. ×0.880 (other)
Region	China		
Net Finished Weight	0.0000 kg	kg	Weight in finished part
Direct Material Cost	£133.56	GBP	Bypasses weight-based model

Data Confidence	Medium
Effective Date	2026-06-14

3B Material Line Items (20 lines)

Ref	Description	Qty	Unit GBP	Ext GBP	Source / basis
U-MCU	Renesas R7F702300B RH850/U2A automotive MCU	1	£15.00	£15.00	
U-ASIC1	Bosch 40342/01 motor-control ASIC (captive)	1	£10.00	£10.00	
U-ASIC2	Bosch 40341/01 ASIC (captive)	1	£7.00	£7.00	
U-ASIC3	Bosch 2302701 ASIC (captive)	1	£4.00	£4.00	
U-SBC	ST L9369 multi-channel driver / SBC	1	£5.00	£5.00	
U-BGA2	Secondary safety MCU / ASIC (BGA)	1	£8.00	£8.00	
U-MEM	External memory (BGA)	1	£3.00	£3.00	
Q1-6	Q142E power stage / dual half-bridge	6	£3.00	£18.00	
U-CAN	NXP TJA1463A CAN-FD SIC transceiver	2	£1.20	£2.40	
U-71H	Bosch 71H740 driver / power-path (captive)	3	£1.50	£4.50	
U-76E2	Bosch 76E240 device (captive)	1	£1.80	£1.80	
U-76E8	Bosch 76E840 device (captive)	1	£1.80	£1.80	
U-7S1R	Bosch 7S1R540H power device (captive)	2	£1.60	£3.20	
C-BULK	Polymer alu cap 150 uF 35 V (351 150 EJV)	6	£0.40	£2.40	
L-PWR	Shielded power inductor 0.47-10 uH	8	£0.60	£4.80	

Ref	Description	Qty	Unit GBP	Ext GBP	Source / basis
R/C-SMD	MLCC + thick-film resistors, AEC-Q200	620	£0.01	£5.58	
D/Q-SML	Small-signal semis (SOT/SOD/QFN)	90	£0.09	£8.10	
Y1	Crystal / oscillator	1	£0.45	£0.45	
J1-2	Automotive press-fit header (main + motor)	2	£2.50	£5.00	
TIM	Thermal interface pad (power stages)	8	£0.12	£0.96	
LINE ITEM TOTAL				£110.99	

Lines total £110.99 against a £133.56 material bucket — the £22.57 balance is carried outside the itemisation (bare board, yield/rework allowance, coating).

§4 — Operations Detail

4A Machine Operations

Operation	Machine Class	Rate / hr	Cycle (min)	Parts / Cycle	OEE %	Machine Cost
SMT Placement & Reflow	High-Speed SMT Line (Fuji/Juki/ASM)	£78.38	5.51	1	82.0%	£8.78
TH Insertion & Hand Soldering	Assembly Workbench	£0.36	0.40	1	100.0 %	£0.00
BGA X-Ray Inspection (100% offline)	X-Ray BGA Inspection Cell (2D/3D AXI)	£47.63	7.92	1	90.0%	£6.99
ICT / Functional Test	ICT Bed-of-Nails Test System (Automotive)	£58.39	4.50	1	95.0%	£4.61
TOTAL						£20.38

4B Labour Detail

Operation	Labour Grade	Rate / hr	Man ning	Lab. min	Eff %	Labour Cost	Op Total
SMT Placement & Reflow	SMT / Electronics Operator	£6.50	1	5.51	100.0%	£0.60	£9.38
TH Insertion & Hand Soldering	SMT / Electronics Operator	£6.50	1	0.40	100.0%	£0.04	£0.05
BGA X-Ray Inspection (100% offline)	SMT / Electronics Operator	£6.50	1	7.92	100.0%	£0.86	£7.84
ICT / Functional Test	SMT / Electronics Operator	£6.50	1	4.50	100.0%	£0.49	£5.10
TOTAL						£1.99	£22.36

\$5 — Machine Rate Buildup

Assembly Workbench [bench-assembly] · Confidence: Low · Assembly bench — low machine rate; cost primarily driven by labour | Regional: capex/overhead ×0.55, energy re-tariffed @£0.07/kWh

Cost Component	Annual Cost (GBP)	Rate/hr @85% util	Notes
Depreciation	£275.00	£0.08	4,000 hr/yr available
Maintenance	£110.00	£0.03	
Energy	£91.30	£0.03	
Floor Space	£440.00	£0.13	
Indirect Support	£275.00	£0.08	
Finance Cost	£27.50	£0.01	
TOTAL — Assembly Workbench	£1218.80	£0.36	Effective hours: 3400/yr

High-Speed SMT Line (Fuji/Juki/ASM) [smt-high-speed-line] · Confidence: Low · High-speed SMT line 80000+ CPH. Automotive EMS benchmark. Target £150/hr. Jun 2026 | Regional: capex/overhead ×0.55, energy re-tariffed @£0.07/kWh

Cost Component	Annual Cost (GBP)	Rate/hr @82% util	Notes
Depreciation	£99000.00	£30.18	4,000 hr/yr available
Maintenance	£44000.00	£13.41	
Energy	£16739.13	£5.10	
Floor Space	£16500.00	£5.03	
Indirect Support	£55000.00	£16.77	
Finance Cost	£25850.00	£7.88	
TOTAL — High-Speed SMT Line (Fuji/Juki/ASM)	£257089.13	£78.38	Effective hours: 3280/yr

X-Ray BGA Inspection Cell (2D/3D AXI) [xray-bga-inspection] · Confidence: Low · 2D/3D automated X-ray inspection for BGA solder joints. EMS automotive benchmark. Target £90/hr. Jun 2026 | Regional: capex/overhead ×0.55, energy re-tariffed @£0.07/kWh

Cost Component	Annual Cost (GBP)	Rate/hr @82% util	Notes
Depreciation	£66000.00	£20.12	4,000 hr/yr available
Maintenance	£24750.00	£7.55	
Energy	£7608.70	£2.32	
Floor Space	£11000.00	£3.35	
Indirect Support	£35750.00	£10.90	
Finance Cost	£11110.00	£3.39	
TOTAL — X-Ray BGA Inspection	£156218.70	£47.63	Effective hours: 3280/yr

ICT Bed-of-Nails Test System (Automotive) [ict-automotive] · Confidence: Low · In-circuit test fixture for IATF 16949 automotive boards. Target £110/hr. Jun 2026 | Regional: capex/overhead ×0.55, energy re-tariffed @£0.07/kWh

Cost Component	Annual Cost (GBP)	Rate/hr @82% util	Notes
Depreciation	£71500.00	£21.80	4,000 hr/yr available
Maintenance	£30250.00	£9.22	
Energy	£9130.43	£2.78	
Floor Space	£13750.00	£4.19	
Indirect Support	£46750.00	£14.25	
Finance Cost	£20130.00	£6.14	
TOTAL — ICT Bed-of-Nails Test System (Automotive)	£191510.43	£58.39	Effective hours: 3280/yr

How to read the machine rate

Rate/hr = (annual depreciation + maintenance + energy + floor + indirect + finance) ÷ effective hours, where effective hours = available hours × utilisation.

Effective hours encode the shift pattern: a lower hours-base raises £/hr and a higher one lowers it. A challenge from a supplier will target this divisor and whether depreciation is machine-only or full-cell (machine + dosing furnace + trim + robots/extraction) — state your basis when defending the number.

§6 — Rate Traceability

Field	Value	Unit	Source / Reference	Rate ID	Conf.
rawMaterial.directCost	133.5586	£	Pre-computed by commodity module	mat-virtual	Medium
SMT Placement & Reflow.machineRatePerHr	78.3808	£/hr	High-speed SMT line 80000+ CPH. Automotive EMS benchmark. Target £150/hr. Jun 2026 Regional: capex/overhead ×0.55, energy re-tariffed @£0.07/kWh	smt-high-speed-line	Low
SMT Placement & Reflow.labourRatePerHr	6.5000	£/hr	Regional benchmark China — 2026 Q2	lab-cn-electronics	Low
TH Insertion & Hand Soldering.machineRatePerHr	0.3585	£/hr	Assembly bench — low machine rate; cost primarily driven by labour Regional: capex/overhead ×0.55, energy re-tariffed @£0.07/kWh	bench-assembly	Low
TH Insertion & Hand Soldering.labourRatePerHr	6.5000	£/hr	Regional benchmark China — 2026 Q2	lab-cn-electronics	Low
BGA X-Ray Inspection (100% offline).machineRatePerHr	47.6277	£/hr	2D/3D automated X-ray inspection for BGA solder joints. EMS automotive benchmark. Target £90/hr. Jun 2026 Regional: capex/overhead ×0.55, energy re-tariffed @£0.07/kWh	xray-bga-inspection	Low
BGA X-Ray Inspection (100% offline).labourRatePerHr	6.5000	£/hr	Regional benchmark China — 2026 Q2	lab-cn-electronics	Low

Field	Value	Unit	Source / Reference	Rate ID	Conf.
ICT / Functional Test.machineRatePerHr	58.3873	£/hr	In-circuit test fixture for IATF 16949 automotive boards. Target £110/hr. Jun 2026 Regional: capex/overhead ×0.55, energy re-tariffed @£0.07/kWh	ict-automotive	Low
ICT / Functional Test.labourRatePerHr	6.5000	£/hr	Regional benchmark China — 2026 Q2	lab-cn-electronics	Low

§7 — Should-Cost with Uncertainty

Monte-Carlo · Low confidence

Estimate (± band)	P10 · optimistic	P50 · median	P90 · conservative	Band	CV
£185.32 ± 23.8%	£144.10	£181.43	£232.19	wide	19%

Range reflects how well each cost driver is known (High / Medium / Low confidence). Tighten it by attaching CAD, a BOM, or logging an actual quote.

Why confidence is Low

Main band drivers: 8 rate(s) at Low confidence; tooling amortisation carries the widest per-bucket spread.

What's excluded from this unit cost

One-time NRE — tooling / die, fixtures and CNC programming — is amortised separately, not in this unit cost.

Import duty and international freight (regional table is Ex-Works).

Formal embodied-carbon reporting (figures are indicative cradle-to-gate — replace with supplier EPDs).

§8 — Sensitivity Analysis

± 10% driver swing · ranked by £ impact

Cost Driver	Base	-10% cost	+10% cost	Range £
Direct material cost	£133.56	-8.5%	+8.5%	£31.45
Overhead %	9%	-0.8%	+0.8%	£3.04
Supplier Margin %	8%	-0.7%	+0.7%	£2.75
SMT Placement & Reflow: Cycle Time	0.09hr	-0.6%	+0.6%	£2.21
SMT Placement & Reflow: Machine Rate	£78.38/hr	-0.6%	+0.6%	£2.07
BGA X-Ray Inspection (100% offline): Cycle Time	0.13hr	-0.5%	+0.5%	£1.85
BGA X-Ray Inspection (100% offline): Machine Rate	£47.63/hr	-0.4%	+0.4%	£1.64
ICT / Functional Test: Cycle Time	0.08hr	-0.3%	+0.3%	£1.20

§9 — Regional Cost Comparison

Ex-Works · per-region should-cost · vs China

Region	Material	Process	Labour	Tooling	Overhead	Ex-Works	Logistics	Total	vs China
United Kingdom (GBP)	£151.77	£37.05	£7.15	£0.56	£18.75	£215.28	£0.39	£230.44	+24%
Germany (EUR)	£156.32	£38.90	£11.55	£0.59	£21.76	£229.13	£0.45	£245.29	+32%
France (EUR)	£154.81	£34.09	£8.30	£0.51	£19.80	£217.51	£0.45	£232.94	+26%
Spain (EUR)	£151.77	£32.60	£5.24	£0.49	£17.23	£207.33	£0.45	£222.07	+20%
Poland (PLN)	£147.22	£26.68	£3.25	£0.40	£14.40	£191.94	£0.47	£205.60	+11%
Turkey (TRY)	£136.59	£22.23	£1.81	£0.33	£11.98	£172.94	£0.51	£185.33	+0%
China (CNY)	£133.56	£20.38	£1.99	£0.31	£14.06	£170.29	£0.57	£185.32	Base
India (INR)	£136.59	£19.27	£1.26	£0.29	£10.81	£168.23	£0.59	£180.23	-3%
Mexico (MXN)	£144.18	£22.23	£2.09	£0.33	£12.56	£181.40	£0.53	£194.29	+5%
United States (USD)	£151.77	£31.49	£9.39	£0.47	£17.50	£210.63	£0.49	£225.55	+22%

Per-region should-cost using regional labour, energy and rate benchmarks. Tooling held fixed. Ex-Works — excludes import duty + international freight. Indicative — confirm with RFQ.

§10 — Embodied Carbon

0.00 kgCO2e · 0.00 /kg · cradle-to-gate

Material	0.00 kgCO2e	Material factor	3.00 kg/kg (Generic (assumed))
Process	0.00 kgCO2e	Grid intensity	0.550 kg/kWh · 0.00 kWh
Logistics	0.00 kgCO2e	Total	0.00 kgCO2e

Indicative cradle-to-gate factors — replace with supplier EPDs for formal reporting.

[CRITICAL] Material is the dominant cost driver

Finding	Raw material at 72.1% of total exceeds the pcba benchmark ceiling of 70%. Material is controlling this part's economics.
Impact	High · up to 1% potential saving
Benchmark	Material %: yours 72.1% vs industry 45–70%
Action 1	Re-cost the BOM at the target volume — line items quoted at prototype price carry the largest error
Action 2	Consolidate part numbers across the passive families (a single 0402 100nF X7R across the board)

[WARNING] Low OEE on "SMT Placement & Reflow"

Finding	OEE of 82% is below the 90% benchmark. Every 5% OEE improvement on this operation saves ~£1.019/part.
Impact	Medium · up to 1% potential saving
Action 1	Implement SMED (Single-Minute Exchange of Die) to reduce changeover losses
Action 2	Schedule preventive maintenance to reduce unplanned downtime

§12 — Cost-Reduction Opportunities (ranked by saving)

4 opportunities · combined £13.16/part (~7%)

Ranked on money per part against this costing. The combined figure is the root-sum-square of the top three, capped at 40% — overlapping actions do not save twice, so the column is deliberately not summed.

#	Category	Action	Why it is on the list	Save/p art	%	Timeframe	Risk
1	Design & Geometry	Design Review for DFM Compliance	Formal DFM review with manufacturing engineering to identify part features that increase cost without functional benefit.	£9.27	5.0%	Quick Win	Low
2	Quality & Inspection	Right-Size Inspection and Test	Inspection/test operations carry 7.0% of part cost as separate steps. Capable processes earn reduced inspection — that is what capability data is for.	£7.41	4.0%	Medium Term	Low
3	Commercial & Sourcing	Raw-Material Price Indexation Clause	With material at 72.1% of cost, the quote embeds the supplier's hedge against metal/resin volatility. An index-linked clause removes that risk premium from the piece price.	£5.56	3.0%	Quick Win	Low
4	Commercial & Sourcing	Annual Volume Re-Commitment for Better Pricing	Provide 12-month rolling volume forecast to supplier to secure better unit pricing and tooling amortisation.	£5.56	3.0%	Quick Win	Low

§13 — Opportunity by Category

3 categories with an actionable lever

Category	Opportunities	Best single action	Best save/part	Category total (indicative)
Design & Geometry	1	Design Review for DFM Compliance	£9.27	£9.27
Quality & Inspection	1	Right-Size Inspection and Test	£7.41	£7.41
Commercial & Sourcing	2	Raw-Material Price Indexation Clause	£5.56	£11.12

§15 — Implementation Roadmap

Quick Wins — Implement Immediately

- Raw-Material Price Indexation Clause
- Design Review for DFM Compliance
- Annual Volume Re-Commitment for Better Pricing